Andres Diaz Aguilar

adiazaguilar@ethz.ch | linkedin.com/in/andresdiazaguilar | Google Scholar

Education

MSc Mechanical Engineering | **ETH Zürich** 09/2024 - current

- GPA: 5.21/6.00. Coursework: Perception and Learning for Robotics, Probabilistic AI, Vision Algorithms for Mobile Robotics, Computational Control, Model Predictive Control, Embedded Control Systems, Introduction to ML

BSc Mechanical Engineering, Cum Laude | **Eindhoven University of Technology (TU/e)** 09/2021 - 08/2024

- Graduated Cum Laude (top 4%) with the Honors Academy distinction
- Thesis: *Through Thickness Microstructure Reconstruction of Polycrystals Using the Phase Field Method*, earning 9.5/10. Results presented at the *2024 ICMS Multiscale Mechanics and Artificial Microstructures Workshop*.

Publications & Presentations

L. Guo*, **A. M. Diaz Aguilar***, W. Talbot, T. Tuna, M. Hutter, C. Cadena. *"CaRLi-V: Camera-RADAR-LiDAR Point-Wise 3D Velocity Estimation."* IEEE Robotics and Automation Practice (RA-P), 2026. doi.org/10.1109/RAP.2026.3702579.

A. M. Diaz Aguilar, A. Lodge, F. Hoekstra. *"Unlocking Electrochemical Battery Modelling: An Open-Source Suite Comparison Methodology."* 11th IFAC Symposium on Advances in Automotive Control, 2025. doi.org/10.1016/j.ifacol.2025.07.100.

A. Lodge, **A. M. Diaz Aguilar**, F. Hoekstra, S. Wilkins. *"Why is PyBaMM so P(y)opular? A Comparative Analysis of Open-Source Battery Modelling Tools."* PyBaMM Battery Modelling Conference 2025, Poster

Reviewer: IEEE ICRA 2027; IFAC AAC 2025.

Relevant Academic Projects

Robotic Systems Lab, ETH Zürich | *ROS2, Python, OpenCV* 11/2025

CaRLi-V: Camera-RADAR-LiDAR Point-Wise 3D Velocity Estimation

- Developed camera–RADAR–LiDAR fusion framework for point-wise dense 3D velocity estimation, leveraging a newly introduced velocity cube RADAR representation for dense radial velocity encoding.
- Proposed method outperforms state-of-the-art scene flow methods in velocity estimation accuracy.
- Published in the *IEEE RA-P journal*, to be presented at *ICRA 2027*. Work also released as open-source *ROS2 package*.

Automatic Control Laboratory, ETH Zürich / Tinamu Labs AG | *ROS, C++, FAST-LIO2* 07/2026

Semester Project: Robust LiDAR-Inertial Odometry in Challenging Sensing Regimes

- Diagnosed FAST-LIO2 failure modes in field deployments, shown to originate from coupled geometric degeneracy, IMU attitude misestimation, and map corruption issues.
- Incorporated LiDAR intensity-augmented registration, motion-aware correspondence filtering, and probabilistic weighting to suppress noise-dominated LiDAR update directions to increase robustness in the observed failure cases.
- Manuscript in preparation for *EEC 2027*.

TU/e Honors Academy (High Tech Systems track) | *ROS2, C++, Python, OpenCV* 02/2024

Mesh-Based Object Detection & Active Perception, supervised by Prof. Elena Torta

- Developed active perception framework for RGB-D-based object detection of objects with known meshes, using point-cloud segmentation and ICP-based mesh alignment to match the mesh to objects in the environment.
- Used detection confidence to select viewpoints that reveal discriminative geometry and resolve ambiguous detections.

Professional Experience

TNO, Netherlands Org. for Applied Scientific Research | *Python, PyBaMM, MATLAB/Simulink* 09/2024 - 02/2025

Battery Modeling and Control Internship

- Developed benchmarking framework for physics-based battery models (PBMs) and performed the first comparative study of leading open-source modeling tools. Presented at *IFAC AAC 2025*, paper published in *proceedings*.
- Developed aging-aware PBM-based fast-charging controller, limiting lithium plating via anode potential constraints.

Robotic Systems Lab Research Assistant, ETH Zürich | *Siemens NX* 08/2026 - current

- Working on robot hardware design and the development of experimental platforms to support the lab's research.

Institute for Dynamic Systems and Control Research Assistant, ETH Zürich | *SolidWorks* 09/2025 - 01/2026

PerfusionX Lab, Prof. Christopher Onder's group

- Worked on the design, fabrication, and assembly of a novel ex-vivo perfusion machine tailored to medical research.

Skills

Programming languages: Python, C/C++, MATLAB/Simulink, Julia
Frameworks & Tools: ROS/ROS2, PyTorch, OpenCV, cvxpy, scikit-learn, Linux, Git
Languages: English (fluent), French (fluent), Spanish (native), Portuguese (native)